PHILIPPINE SOCIETY OF
ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition
for a Competitive and Sustainable Future

6 OCTOBER 2026
Grand Ballroom,
Okada Manila










PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

Management & Feeding Strategies in Modern Tilapia Farming

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Outline of Presentation

I. Understanding Modern Tilapia Farming
II. Biomass Management
III. Feeding Strategies
IV. FCR & Feed Efficiency
V. Feeding vs Water Quality
VI. Precision Feeding
VII. Economics
VIII. Case Study (Sual case)
IX. Conclusion

I. UNDERSTANDING MODERN TILAPIA FARMING



I. UNDERSTANDING MODERN TILAPIA FARMING



II. Biomass Management: Stocking Density



II. Biomass Management: Biomass - The Foundation of Feeding

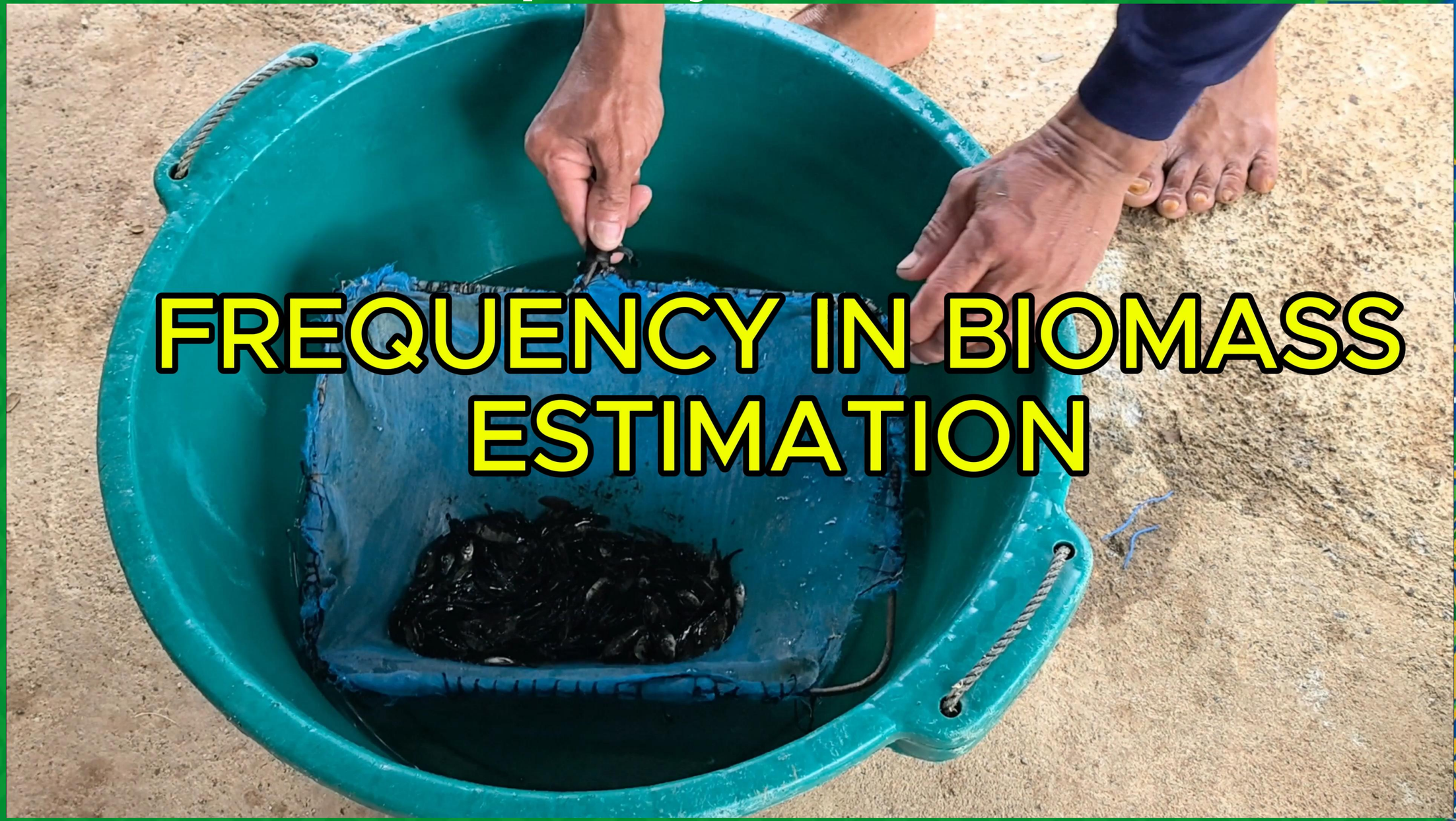
- Computing Biomass
- Compute feed consumption/day baked on feeding rate

10,000 fish × 100 g = 1,000 kg biomass

Feeding rate 2%

1,000 kg × 2% = 20 kg feed/day

II. Biomass Management: ***Frequency of Biomass Estimation***



II. Biomass Management: Most Practical Way of Biomass Estimation

- *MOST practical way - Feeding at Satiation*

Amount fed at satiation	**Assumed feeding rate**	**Estimated Biomass**
100kgs	2%	5,000 kgs
	3%	3,333 kgs
	4%	2,500 kgs

Estimated Biomass = 100kgs/**Assumed FR**

III. FEEDING STRATEGIES



III. Feeding Strategies: Feeding Rate: Not a Fixed Number



III. Feeding Strategies: Calculating Daily Feed Req't

- Fish population
- ABW
- Feeding rate
- Feeding frequency

Fish population: 20,000

Average weight: 50 g

Biomass:

20,000 × 0.05 kg = **1,000 kg**

At **3% feeding rate**:

1,000 × 0.03 = **30 kg feed/day**

If divided into 3 meals:

10 kg/feeding

III. Feeding Strategies: Feeding frequency & Feed Utilization

- Feeding frequency: 1x vs. 2-3x feeding/day
  - Feed Intake
  - Digestion
  - Growth
- Feed wastage
- Labor
- Automation

More frequent feeding doe not automatically mean better FCR:
The quality of feeding reponse matter

III. Feeding Strategies: Appetite-based Feeding

- Feed → Observe → Adjust
- Monitor:
  - Feeding activity
  - Surface behavior
  - Feed disappearance
  - Uneaten pellets
  - Fish distribution
  - Environmental Conditions

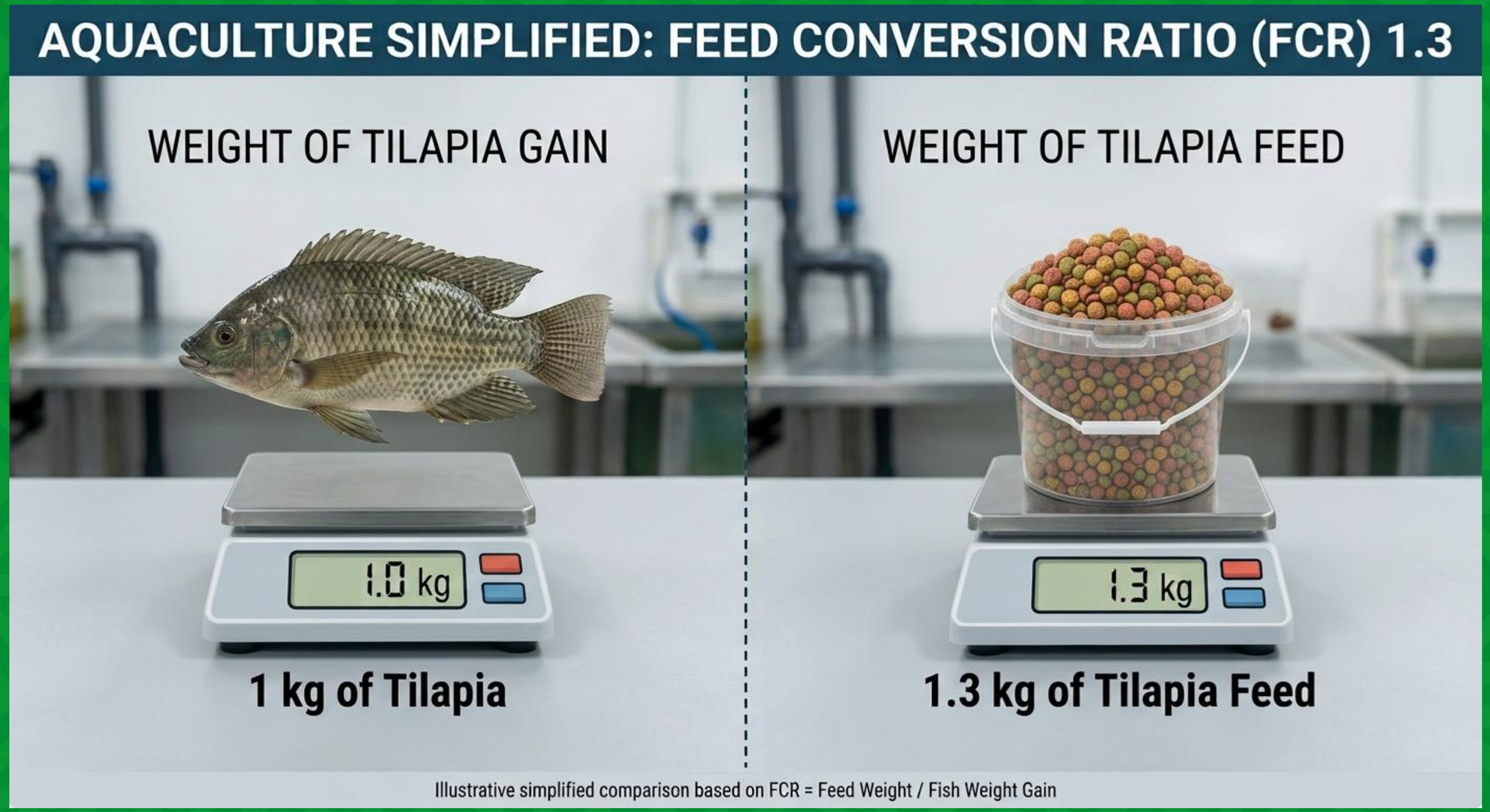
III. Feeding Strategies: When to reduce or stop feeding?



IV. FCR & FEED EFFICIENCY



IV. FCR & Feed Efficiency: Understanding FCR



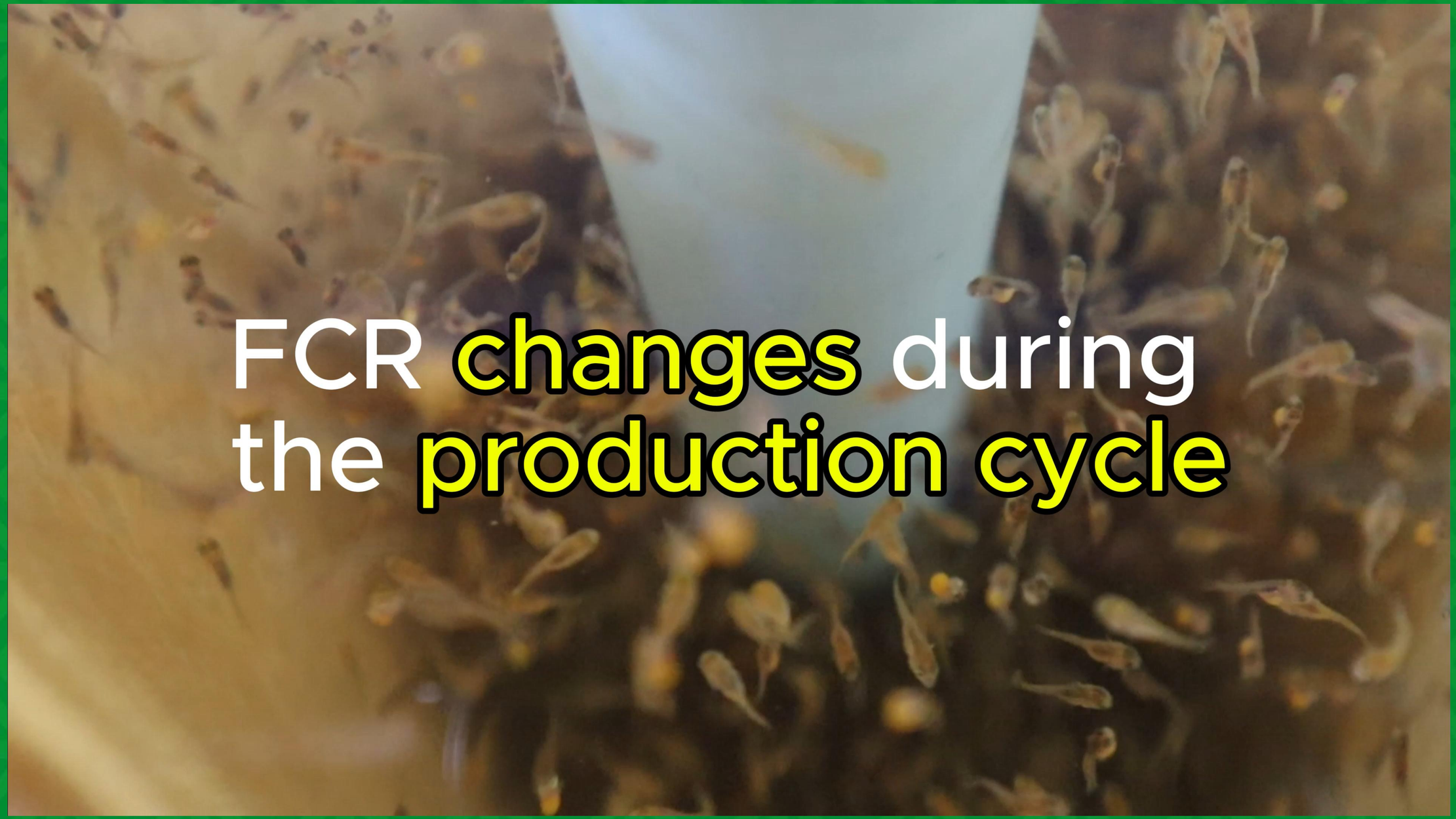
IV. FCR & Feed Efficiency: What determines FCR

- Genetics | Feed |
- Environment | Fish size
- Health | Feeding strategy
- Stocking density | DO | Temperature

FCR is NOT merely a feed problem,
It is a farm-management indicator

DO: Dissolved oxygen

IV. FCR & Feed Efficiency: Why FCR changes during the production cycle

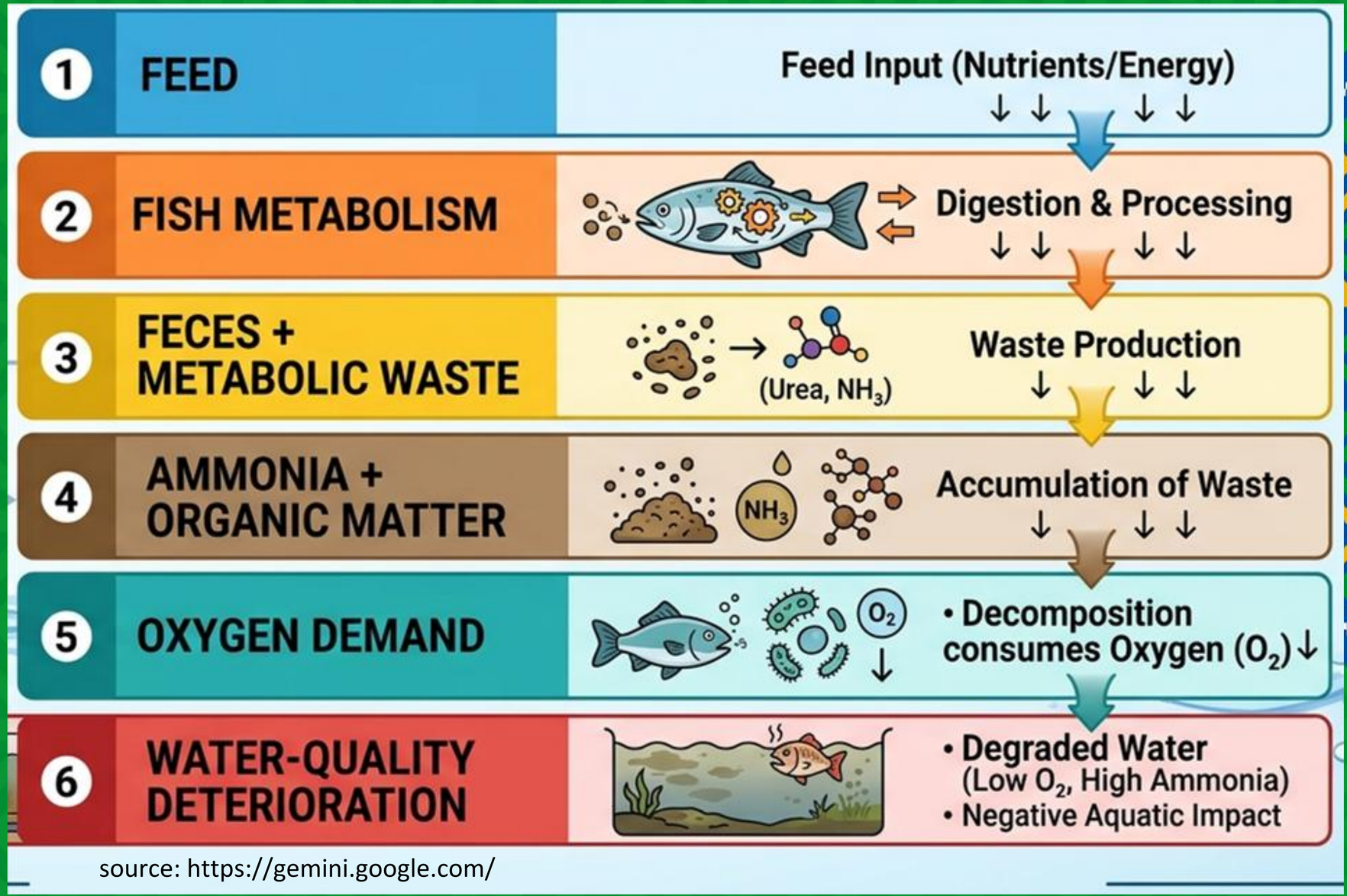


IV. FCR & Feed Efficiency: Why FCR changes during the production cycle



Harvest timing matters

V. FEEDING vs WATER QUALITY:



source: https://gemini.google.com/

V. Feeding vs Water Quality: Dissolved oxygen: The hidden feeding variable

Low DO

- → reduced appetite
- → reduced growth
- → poorer feed utilization

DO: Dissolved oxygen

V. Feeding vs Water Quality: Dissolved oxygen: The hidden feeding variable

At HIGH Biomass:

- **More** feeds
- → **More** metabolic activity + decomposition
- → **Higher** oxygen demand

Aeration capacity mut be matched with feeding and biomass

V. Feeding vs Water Quality: Temperature, pH and ammonia

Temperature affects:

- Metabolism
- Appetite
- Digestion
- Growth

V. Feeding vs Water Quality: Temperature, pH and ammonia

***pH** affects:*

- **proportion of toxic**

V. Feeding vs Water Quality: Temperature, pH and ammonia

Ammonia:

- **TAN =** Unionized **NH3 +** Ionized **NH4**

V. Feeding vs Water Quality: Temperature, pH and ammonia

AMMONIA TOXICITY EQUILIBRIUM IN AQUACULTURE

Effect of pH and Temperature on Unionized Ammonia (NH_3)

'As pH and temperature increase, the proportion of TAN present as toxic unionized NH_3 increases.'

TOTAL AMMONIUM NITROGEN (TAN) EQUILIBRIUM

AMMONIUM ION (NH_4^+) ⟷ UNIONIZED AMMONIA (NH_3)

pH & pH & Temperature

Less Toxic — Toxic

FISH HEALTH (Safe) — FISH HEALTH (Toxicity Risk)

$$TAN = [NH_4^+] + [NH_3]$$

*Values are illustrative for common aquaculture conditions.

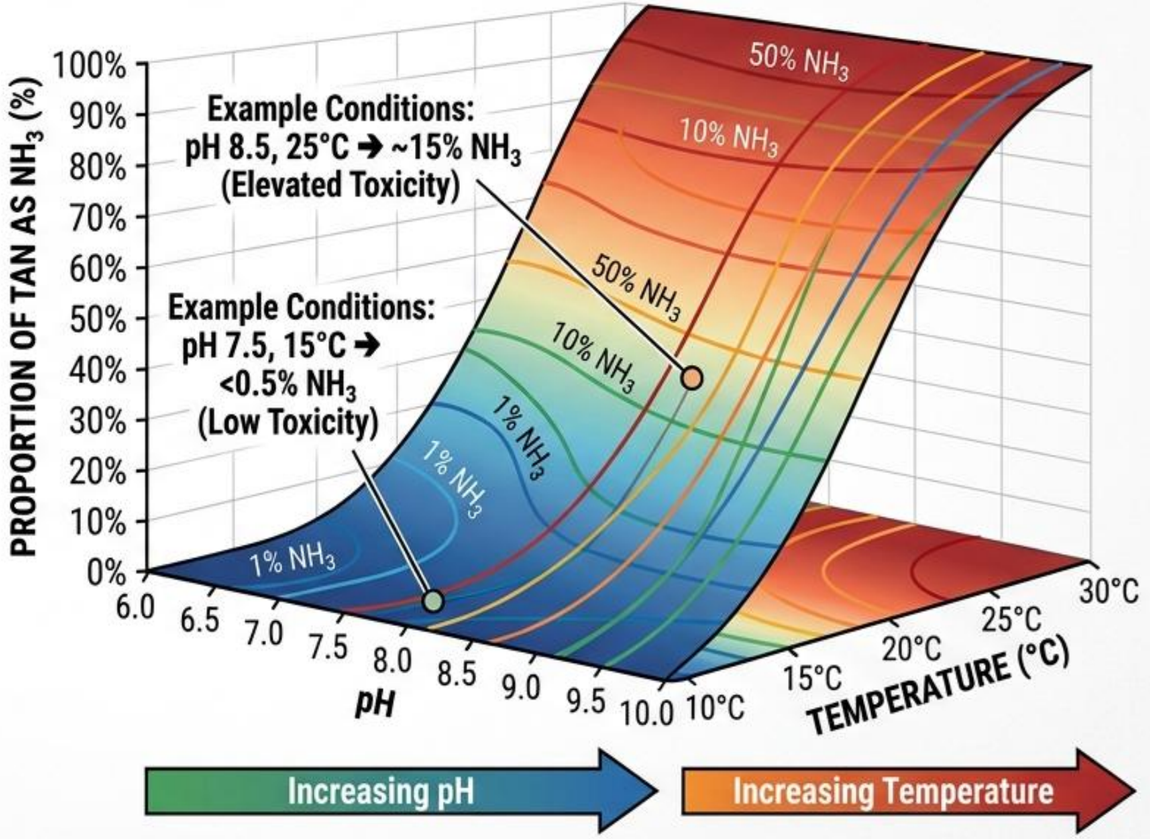


FIG. 1

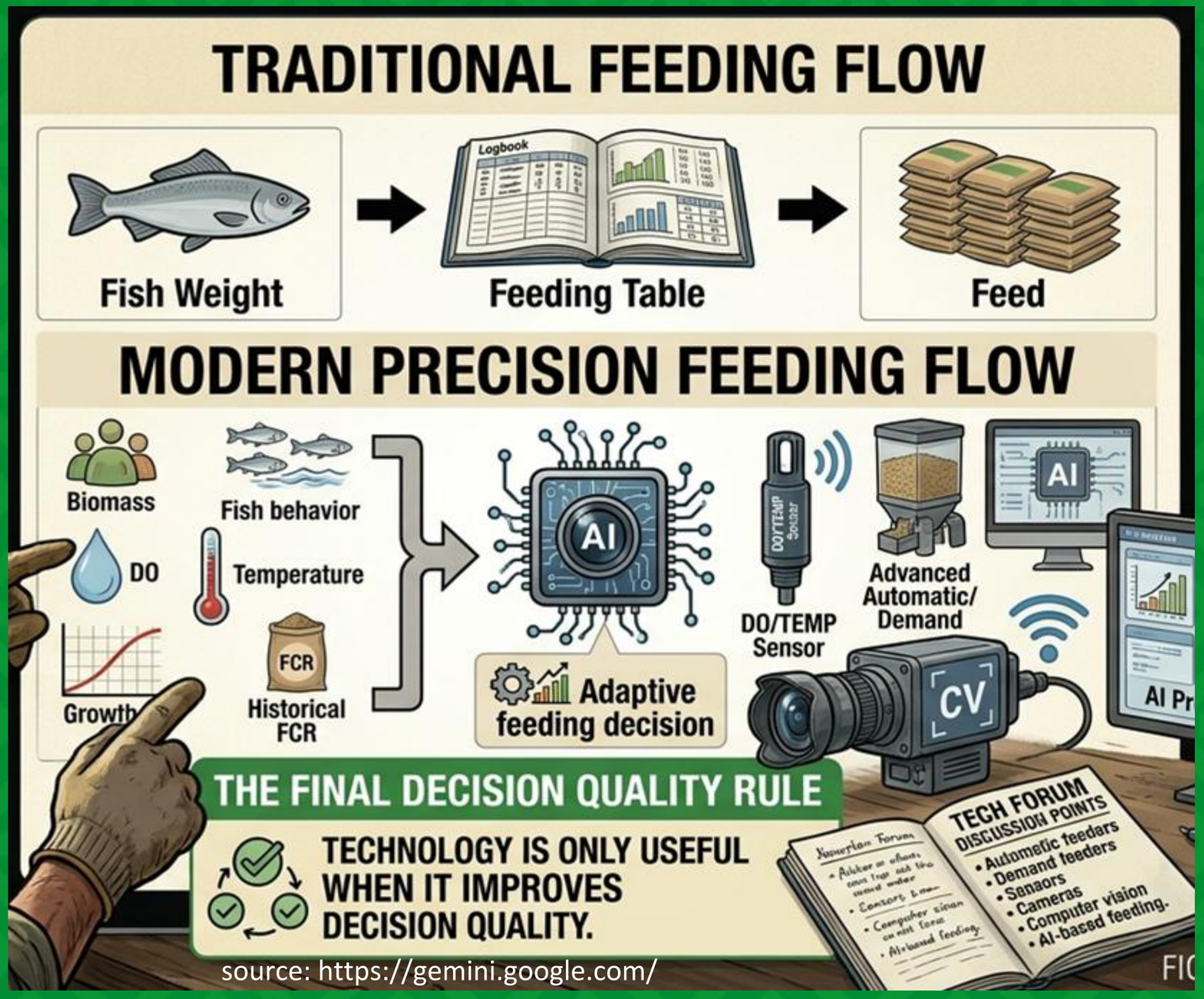
source: https://gemini.google.com/

Higher feeding → greater nitrogen input → greater ammonia production potential.

VI. PRECISION FEEDING

1. From feeding tables to precision feeding
2. Digital farm monitoring

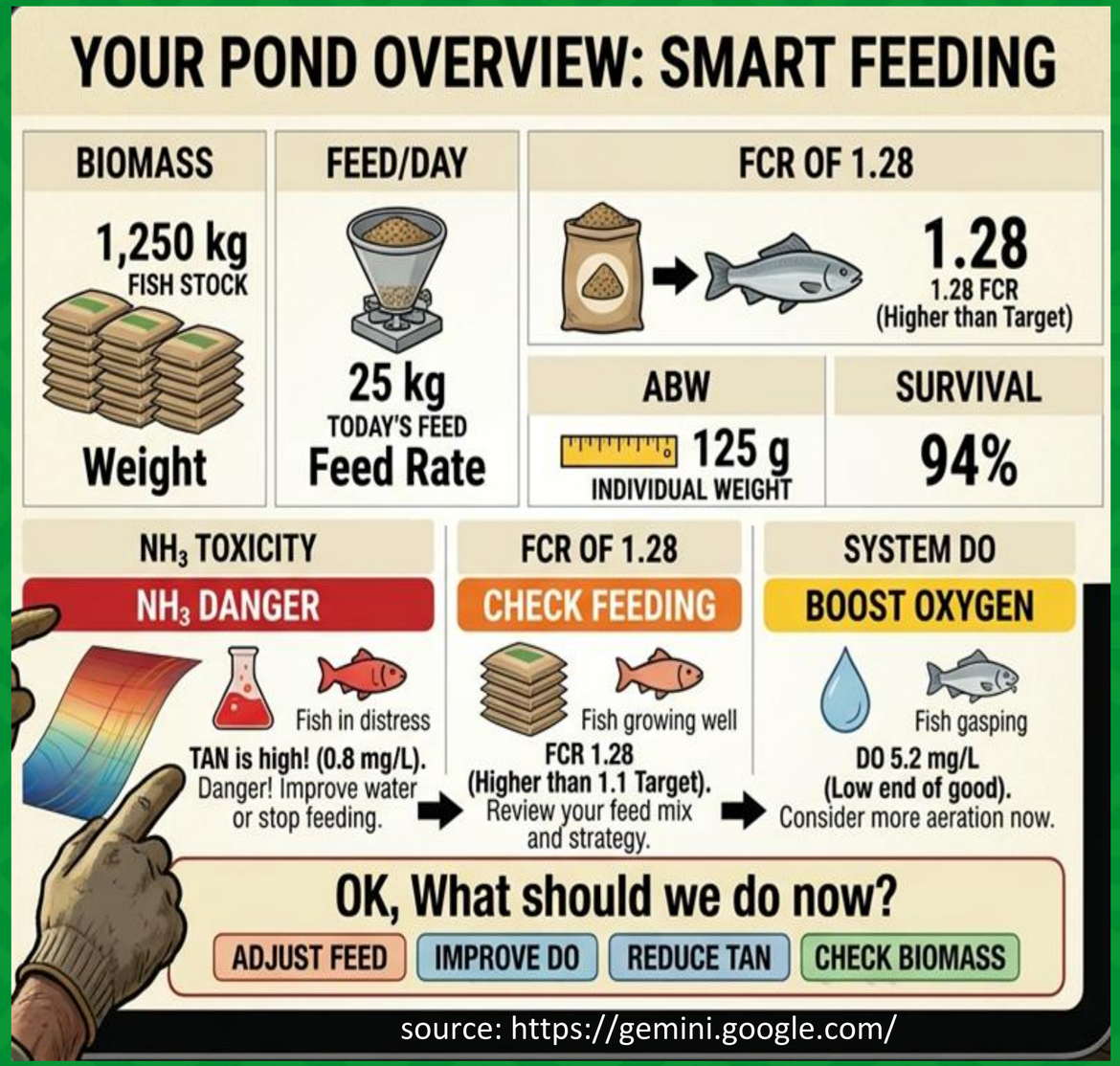
VI. Precision Feeding: From Feeding Tables to Precision Feeding



source: https://gemini.google.com/



VI. Precision Feeding: Digital Farm Monitoring



source: https://gemini.google.com/



VII. Economics: FCR is Money



VIII. Case Study: Stress vs Carrying Capacity

2004-2009





Stocking: 50,000 pcs/cage



CC: 50-60 kgs/m3

2010-2019





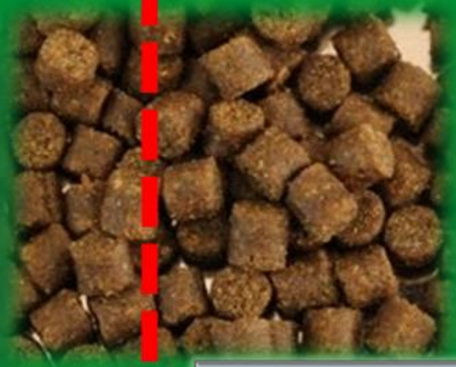


CC: 11-19 kgs/m3

VIII. Case Study: Stress vs FCR

20 yrs ago





Present



Less ENERGY

REGULAR

- Respiration
- Osmoregulation
- Immunity
- Waste excretion

FCR = 2.0

GROWTH

Less ENERGY

IRREGULAR

- Respiration
- Osmoregulation
- Immunity
- Waste excretion
- STRESS exposure
- SURVIVAL mode

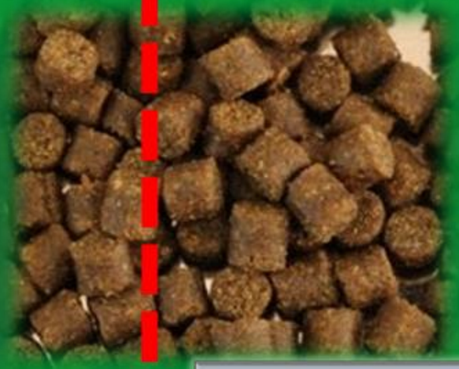
GROWTH

FCR = 2.9

VIII. Case Study: Stress vs FCR

20 yrs ago





Present



Less ENERGY

REGULAR

- Respiration
- Osmoregulation
- Immunity
- Waste excretion

FCR = 2.0

GROWTH

Less ENERGY

IRREGULAR

- Respiration
- Osmoregulation
- Immunity
- Waste excretion
- STRESS exposure
- SURVIVAL mode

GROWTH

FCR = 2.9

VIII. Case Study: Stress vs Survival Rate

20 yrs ago



REGULAR

- Good water
- Carrying capacity
- Immunity
- Waste excretion

SURVIVAL

50-90%

Present



IRREGULAR

- Bad water
- Carrying capacity
- Immunity
- Waste excretion
- STRESS exposure
- SURVIVAL mode

SURVIVAL

20-60%

VIII. Case Study: Stress vs DWG

20 yrs ago



REGULAR

- Good water
- Carrying capacity
- Immunity
- Waste excretion

DWG

6-9g/day

Present



IRREGULAR

- Bad water
- Carrying capacity
- Immunity
- Waste excretion
- STRESS exposure
- SURVIVAL mode

DWG

2-4 g/day

VIII. Case Study: Stress vs DOC

20 yrs ago



REGULAR

- Good water
- Carrying capacity
- Immunity
- Waste excretion

DOC

4-5 mos.

Present



IRREGULAR

- Bad water
- Carrying capacity
- Immunity
- Waste excretion
- STRESS exposure
- SURVIVAL mode

DOC

6-9 mos.

IX. CONCLUSION



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Thank you!

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